平方零厚化上导出复形的障碍消失判据

Counting scope. The primary problem is Tag 0DIW, marked below. All preceding numbered statements are supporting author context and are not counted as additional problems. This is a local pilot with a temporary ID, not production admission.

Copyright (C) 2005–2025 Johan de Jong. Stacks Project authors. Source: Tag 0DIW.

Difficulty H2: 本题证明导出复形提升存在当且仅当其规范二次障碍类为零。作者把零同伦作为校正项在纤维积模上定义新的微分，再通过导出张量、上同调下降归纳和四引理证明模掉平方零理想后得到原复形。规范类的构造可以明确作为背景，但校正复形与拟同构验证不能由背景拿掉，判 H2。.

History: excerpt from revision local cached snapshot; exact chapter hashes; revision unknown, defos.tex, lines 5357–5495. Only reference presentation was adapted. Licensed under GNU FDL 1.2 or later; no invariant sections or cover texts. See accompanying COPYING.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{O} \to \mathcal{O}_0$ be a surjection of sheaves of rings whose kernel is an ideal sheaf $\mathcal{I}$ of square zero. For every object K_0 in $D^-(\mathcal{O}_0)$ there is a canonical map*

$$\omega(K_0) : K_0 \longrightarrow K_0 \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I}[2]$$

in $D(\mathcal{O}_0)$ such that for any map $K_0 \to L_0$ in $D^-(\mathcal{O}_0)$ the diagram

$$\begin{array}{ccc} K_0 & \xrightarrow{\omega(K_0)} & (K_0 \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I})[2] \\ \downarrow & & \downarrow \\ L_0 & \xrightarrow{\omega(L_0)} & (L_0 \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I})[2] \end{array}$$

commutes.

证明. Represent K_0 by any complex $\mathcal{K}_0^\bullet$ of $\mathcal{O}_0$-modules. Apply Lemma 0.2 with $\mathcal{G}^n = 0$ for all n. Denote $d : \mathcal{F}^n \to \mathcal{F}^{n+1}$ the maps produced by the lemma. Then we see that $d \circ d : \mathcal{F}^n \to \mathcal{F}^{n+2}$ is zero modulo $\mathcal{I}$. Since $\mathcal{F}^n$ is flat, we see that $\mathcal{I}\mathcal{F}^n = \mathcal{F}^n \otimes_{\mathcal{O}} \mathcal{I} = \mathcal{F}_0^n \otimes_{\mathcal{O}_0} \mathcal{I}$. Hence we obtain a canonical map of complexes

$$d \circ d : \mathcal{F}_0^\bullet \longrightarrow (\mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I})[2]$$

Since $\mathcal{F}_0^\bullet$ is a bounded above complex of flat $\mathcal{O}_0$-modules, it is K-flat and may be used to compute derived tensor product. Moreover, the map of complexes $\mathcal{F}_0^\bullet \to \mathcal{K}_0^\bullet$ is a quasi-isomorphism by construction. Therefore the source and target of the map just constructed represent K_0 and $K_0 \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I}[2]$ and we obtain our map $\omega(K_0)$.

Let us show that this procedure is compatible with maps of complexes. Namely, let $\mathcal{L}_0^\bullet$ represent another object of $D^-(\mathcal{O}_0)$ and suppose that

$$\mathcal{K}_0^\bullet \longrightarrow \mathcal{L}_0^\bullet$$

is a map of complexes. Apply Lemma 0.2 for the complex $\mathcal{L}_0^\bullet$, the flat modules $\mathcal{F}^n$, the maps $\mathcal{F}^n \to \mathcal{F}^{n+1}$, and the compositions $\mathcal{F}^n \to \mathcal{K}_0^n \to \mathcal{L}_0^n$ (we apologize for the reversal of letters used).

We obtain flat modules $\mathcal{G}^n$, maps $\mathcal{F}^n \to \mathcal{G}^n$, maps $\mathcal{G}^n \to \mathcal{G}^{n+1}$, and maps $\mathcal{G}^n \to \mathcal{L}_0^n$ with all properties as in the lemma. Then it is clear that

$$\begin{array}{ccc} \mathcal{F}_0^\bullet & \longrightarrow & (\mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I})[2] \\ \downarrow & & \downarrow \\ \mathcal{G}_0^\bullet & \longrightarrow & (\mathcal{G}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I})[2] \end{array}$$

is a commutative diagram of complexes.

To see that $\omega(K_0)$ is well defined, suppose that we have two complexes $\mathcal{K}_0^\bullet$ and $(\mathcal{K}_0')^\bullet$ of $\mathcal{O}_0$-modules representing K_0 and two systems $(\mathcal{F}^n, d : \mathcal{F}^n \to \mathcal{F}^{n+1}, \mathcal{F}^n \to \mathcal{K}_0^n)$ and $((\mathcal{F}')^n, d : (\mathcal{F}')^n \to (\mathcal{F}')^{n+1}, (\mathcal{F}')^n \to \mathcal{K}_0^n)$ as above. Then we can choose a complex $(\mathcal{K}_0'')^\bullet$ and quasi-isomorphisms $\mathcal{K}_0^\bullet \to (\mathcal{K}_0'')^\bullet$ and $(\mathcal{K}_0')^\bullet \to (\mathcal{K}_0'')^\bullet$ realizing the fact that both complexes represent K_0 in the derived category. Next, we apply the result of the previous paragraph to

$$(\mathcal{K}_0)^\bullet \oplus (\mathcal{K}_0')^\bullet \longrightarrow (\mathcal{K}_0'')^\bullet$$

This produces a commutative diagram

$$\begin{array}{ccc} \mathcal{F}_0^\bullet \oplus (\mathcal{F}_0')^\bullet & \longrightarrow & (\mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I})[2] \oplus ((\mathcal{F}_0')^\bullet \otimes_{\mathcal{O}_0} \mathcal{I})[2] \\ \downarrow & & \downarrow \\ \mathcal{G}_0^\bullet & \longrightarrow & (\mathcal{G}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I})[2] \end{array}$$

Since the vertical arrows give quasi-isomorphisms on the summands we conclude the desired commutativity in $D(\mathcal{O}_0)$.

Having established well-definedness, the statement on compatibility with maps is a consequence of the result in the second paragraph. □

Lemma 0.2. *Let $\mathcal{C}$ be a site. Let $\mathcal{O} \to \mathcal{O}_0$ be a surjection of sheaves of rings. Assume given the following data*

(1) *flat $\mathcal{O}$-modules $\mathcal{G}^n$,*
(2) *maps of $\mathcal{O}$-modules $\mathcal{G}^n \to \mathcal{G}^{n+1}$,*
(3) *a complex $\mathcal{K}_0^\bullet$ of $\mathcal{O}_0$-modules,*
(4) *maps of $\mathcal{O}$-modules $\mathcal{G}^n \to \mathcal{K}_0^n$*

such that

(a) *$H^n(\mathcal{K}_0^\bullet) = 0$ for $n \gg 0$,*
(b) *$\mathcal{G}^n = 0$ for $n \gg 0$,*
(c) *with $\mathcal{G}_0^n = \mathcal{G}^n \otimes_{\mathcal{O}} \mathcal{O}_0$ the induced maps determine a complex $\mathcal{G}_0^\bullet$ and a map of complexes $\mathcal{G}_0^\bullet \to \mathcal{K}_0^\bullet$.*

Then there exist

(i) *flat $\mathcal{O}$-modules $\mathcal{F}^n$,*
(ii) *maps of $\mathcal{O}$-modules $\mathcal{F}^n \to \mathcal{F}^{n+1}$,*
(iii) *maps of $\mathcal{O}$-modules $\mathcal{F}^n \to \mathcal{K}_0^n$,*
(iv) *maps of $\mathcal{O}$-modules $\mathcal{G}^n \to \mathcal{F}^n$,*

such that $\mathcal{F}^n = 0$ for $n \gg 0$, such that the diagrams

$$\begin{array}{ccc} \mathcal{G}^n & \longrightarrow & \mathcal{G}^{n+1} \\ \downarrow & & \downarrow \\ \mathcal{F}^n & \longrightarrow & \mathcal{F}^{n+1} \end{array}$$

commute for all n, such that the composition $\mathcal{G}^n \to \mathcal{F}^n \to \mathcal{K}_0^n$ is the given map $\mathcal{G}^n \to \mathcal{K}_0^n$, and such that with $\mathcal{F}_0^n = \mathcal{F}^n \otimes_{\mathcal{O}} \mathcal{O}_0$ we obtain a complex $\mathcal{F}_0^\bullet$ and map of complexes $\mathcal{F}_0^\bullet \to \mathcal{K}_0^\bullet$ which is a quasi-isomorphism.

证明. We will prove by descending induction on e that we can find $\mathcal{F}^n$, $\mathcal{G}^n \to \mathcal{F}^n$, and $\mathcal{F}^n \to \mathcal{F}^{n+1}$ for $n \geq e$ fitting into a commutative diagram

$$\begin{array}{ccccccccc} \ldots & \longrightarrow & \mathcal{G}^{e-1} & \longrightarrow & \mathcal{G}^e & \longrightarrow & \mathcal{G}^{e+1} & \longrightarrow & \ldots \\ & & & & \downarrow & & \downarrow & & \\ & & & & \mathcal{F}^e & \longrightarrow & \mathcal{F}^{e+1} & \longrightarrow & \ldots \\ & & & & \downarrow & & \downarrow & & \\ \ldots & \longrightarrow & \mathcal{K}_0^{e-1} & \longrightarrow & \mathcal{K}_0^e & \longrightarrow & \mathcal{K}_0^{e+1} & \longrightarrow & \ldots \end{array}$$

such that $\mathcal{F}_0^\bullet$ is a complex, the induced map $\mathcal{F}_0^\bullet \to \mathcal{K}_0^\bullet$ induces an isomorphism on H^n for $n > e$ and a surjection for $n = e$. For $e \gg 0$ this is true because we can take $\mathcal{F}^n = 0$ for $n \geq e$ in that case by assumptions (a) and (b).

Induction step. We have to construct $\mathcal{F}^{e-1}$ and the maps $\mathcal{G}^{e-1} \to \mathcal{F}^{e-1}$, $\mathcal{F}^{e-1} \to \mathcal{F}^e$, and $\mathcal{F}^{e-1} \to \mathcal{K}_0^{e-1}$. We will choose $\mathcal{F}^{e-1} = A \oplus B \oplus C$ as a direct sum of three pieces.

For the first we take $A = \mathcal{G}^{e-1}$ and we choose our map $\mathcal{G}^{e-1} \to \mathcal{F}^{e-1}$ to be the inclusion of the first summand. The maps $A \to \mathcal{K}_0^{e-1}$ and $A \to \mathcal{F}^e$ will be the obvious ones.

To choose B we consider the surjection (by induction hypothesis)

$$\gamma : \mathrm{Ker}(\mathcal{F}_0^e \to \mathcal{F}_0^{e+1}) \longrightarrow \mathrm{Ker}(\mathcal{K}_0^e \to \mathcal{K}_0^{e+1}) / \mathrm{Im}(\mathcal{K}_0^{e-1} \to \mathcal{K}_0^e)$$

We can choose a set I, for each $i \in I$ an object U_i of $\mathcal{C}$, and sections $s_i \in \mathcal{F}^e(U_i)$, $t_i \in \mathcal{K}_0^{e-1}(U_i)$ such that

(1) s_i maps to a section of $\mathrm{Ker}(\gamma) \subset \mathrm{Ker}(\mathcal{F}_0^e \to \mathcal{F}_0^{e+1})$,
(2) s_i and t_i map to the same section of $\mathcal{K}_0^e$,
(3) the sections s_i generate $\mathrm{Ker}(\gamma)$ as an $\mathcal{O}_0$-module.

We omit giving the full justification for this; one uses that $\mathcal{F}^e \to \mathcal{F}_0^e$ is a surjective maps of sheaves of sets. Then we set to put

$$B = \bigoplus\nolimits_{i \in I} j_{U_i!}\mathcal{O}_{U_i}$$

and define the maps $B \to \mathcal{F}^e$ and $B \to \mathcal{K}_0^{e-1}$ by using s_i and t_i to determine where to send the summand $j_{U_i!}\mathcal{O}_{U_i}$.

With $\mathcal{F}^{e-1} = A \oplus B$ and maps as above, this produces a diagram as above for $e - 1$ such that $\mathcal{F}_0^\bullet \to \mathcal{K}_0^\bullet$ induces an isomorphism on H^n for $n \geq e$. To get the map to be surjective on H^{e-1} we choose the summand C as follows. Choose a set J, for each $j \in J$ an object U_j of $\mathcal{C}$ and a section t_j of $\mathrm{Ker}(\mathcal{K}_0^{e-1} \to \mathcal{K}_0^e)$ over U_j such that these sections generate this kernel over $\mathcal{O}_0$. Then we put

$$C = \bigoplus\nolimits_{j \in J} j_{U_j!}\mathcal{O}_{U_j}$$

and the zero map $C \to \mathcal{F}^e$ and the map $C \to \mathcal{K}_0^{e-1}$ by using s_j to determine where to the summand $j_{U_j!}\mathcal{O}_{U_j}$. This finishes the induction step by taking $\mathcal{F}^{e-1} = A \oplus B \oplus C$ and maps as indicated. □

Lemma 0.3. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\alpha : K \to L$ be a map of $D^-(\mathcal{O})$. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}$-modules. Let $n \in \mathbf{Z}$.*

(1) *If $H^i(\alpha)$ is an isomorphism for $i \geq n$, then $H^i(\alpha \otimes^{\mathbf{L}}_{\mathcal{O}} id_{\mathcal{F}})$ is an isomorphism for $i \geq n$.*
(2) *If $H^i(\alpha)$ is an isomorphism for $i > n$ and surjective for $i = n$, then $H^i(\alpha \otimes^{\mathbf{L}}_{\mathcal{O}} id_{\mathcal{F}})$ is an isomorphism for $i > n$ and surjective for $i = n$.*

Primary problem: Tag 0DIW

Lemma 0.4. *Let $\mathcal{C}$ be a site. Let $\mathcal{O} \to \mathcal{O}_0$ be a surjection of sheaves of rings whose kernel is an ideal sheaf $\mathcal{I}$ of square zero. For every object K_0 in $D^-(\mathcal{O}_0)$ the following are equivalent*

(1) *the class $\omega(K_0) \in \mathrm{Ext}^2_{\mathcal{O}_0}(K_0, K_0 \otimes_{\mathcal{O}_0} \mathcal{I})$ constructed in Lemma 0.1 is zero,*
(2) *there exists $K \in D^-(\mathcal{O})$ with $K \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{O}_0 = K_0$ in $D(\mathcal{O}_0)$.*

证明. Let K be as in (2). Then we can represent K by a bounded above complex $\mathcal{F}^\bullet$ of flat $\mathcal{O}$-modules. Then $\mathcal{F}_0^\bullet = \mathcal{F}^\bullet \otimes_{\mathcal{O}} \mathcal{O}_0$ represents K_0 in $D(\mathcal{O}_0)$. Since $d_{\mathcal{F}^\bullet} \circ d_{\mathcal{F}^\bullet} = 0$ as $\mathcal{F}^\bullet$ is a complex, we see from the very construction of $\omega(K_0)$ that it is zero.

Assume (1). Let $\mathcal{F}^n$, $d : \mathcal{F}^n \to \mathcal{F}^{n+1}$ be as in the construction of $\omega(K_0)$. The nullity of $\omega(K_0)$ implies that the map

$$\omega = d \circ d : \mathcal{F}_0^\bullet \longrightarrow (\mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I})[2]$$

is zero in $D(\mathcal{O}_0)$. By definition of the derived category as the localization of the homotopy category of complexes of $\mathcal{O}_0$-modules, there exists a quasi-isomorphism $\alpha : \mathcal{G}_0^\bullet \to \mathcal{F}_0^\bullet$ such that there exist $\mathcal{O}_0$-modules maps $h^n : \mathcal{G}_0^n \to \mathcal{F}_0^{n+1} \otimes_{\mathcal{O}} \mathcal{I}$ with

$$\omega \circ \alpha = d_{\mathcal{F}_0^\bullet \otimes \mathcal{I}} \circ h + h \circ d_{\mathcal{G}_0^\bullet}$$

We set

$$\mathcal{H}^n = \mathcal{F}^n \times_{\mathcal{F}_0^n} \mathcal{G}_0^n$$

and we define

$$d' : \mathcal{H}^n \longrightarrow \mathcal{H}^{n+1}, \quad (f^n, g_0^n) \longmapsto (d(f^n) - h^n(g_0^n), d(g_0^n))$$

with obvious notation using that $\mathcal{F}_0^{n+1} \otimes_{\mathcal{O}_0} \mathcal{I} = \mathcal{F}^{n+1} \otimes_{\mathcal{O}} \mathcal{I} = \mathcal{I}\mathcal{F}^{n+1} \subset \mathcal{F}^{n+1}$. Then one checks $d' \circ d' = 0$ by our choice of h^n and definition of ω. Hence $\mathcal{H}^\bullet$ defines an object in $D(\mathcal{O})$. On the other hand, there is a short exact sequence of complexes of $\mathcal{O}$-modules

$$0 \to \mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I} \to \mathcal{H}^\bullet \to \mathcal{G}_0^\bullet \to 0$$

We still have to show that $\mathcal{H}^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{O}_0$ is isomorphic to K_0. Choose a quasi-isomorphism $\mathcal{E}^\bullet \to \mathcal{H}^\bullet$ where $\mathcal{E}^\bullet$ is a bounded above complex of flat $\mathcal{O}$-modules. We obtain a commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathcal{E}^\bullet \otimes_{\mathcal{O}} \mathcal{I} & \longrightarrow & \mathcal{E}^\bullet & \longrightarrow & \mathcal{E}_0^\bullet & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle \delta} & & \\
0 & \longrightarrow & \mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I} & \longrightarrow & \mathcal{H}^\bullet & \longrightarrow & \mathcal{G}_0^\bullet & \longrightarrow & 0
\end{array}$$

We claim that δ is a quasi-isomorphism. Since $H^i(\delta)$ is an isomorphism for $i \gg 0$, we can use descending induction on n such that $H^i(\delta)$ is an isomorphism for $i \geq n$. Observe that $\mathcal{E}^\bullet \otimes_{\mathcal{O}} \mathcal{I}$

represents $\mathcal{E}_0^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I}$, that $\mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I}$ represents $\mathcal{G}_0^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I}$, and that $\beta = \delta \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathrm{id}_{\mathcal{I}}$ as maps in $D(\mathcal{O}_0)$. This is true because $\beta = (\alpha \otimes \mathrm{id}_{\mathcal{I}}) \circ (\delta \otimes \mathrm{id}_{\mathcal{I}})$. Suppose that $H^i(\delta)$ is an isomorphism in degrees $\geq n$. Then the same is true for β by what we just said and Lemma 0.3. Then we can look at the diagram

$$\begin{array}{ccccccccc}
H^{n-1}(\mathcal{E}^\bullet \otimes_{\mathcal{O}} \mathcal{I}) & \longrightarrow & H^{n-1}(\mathcal{E}^\bullet) & \longrightarrow & H^{n-1}(\mathcal{E}_0^\bullet) & \longrightarrow & H^n(\mathcal{E}^\bullet \otimes_{\mathcal{O}} \mathcal{I}) & \longrightarrow & H^n(\mathcal{E}^\bullet) \\
\downarrow{\scriptstyle H^{n-1}(\beta)} & & \downarrow & & \downarrow{\scriptstyle H^{n-1}(\delta)} & & \downarrow{\scriptstyle H^n(\beta)} & & \downarrow \\
H^{n-1}(\mathcal{F}_0^\bullet \otimes_{\mathcal{O}} \mathcal{I}) & \longrightarrow & H^{n-1}(\mathcal{H}^\bullet) & \longrightarrow & H^{n-1}(\mathcal{G}_0^\bullet) & \longrightarrow & H^n(\mathcal{F}_0^\bullet \otimes_{\mathcal{O}} \mathcal{I}) & \longrightarrow & H^n(\mathcal{H}^\bullet)
\end{array}$$

Using Homology, Lemma lemma four lemma (Tag 05QA) we see that $H^{n-1}(\delta)$ is surjective. This in turn implies that $H^{n-1}(\beta)$ is surjective by Lemma 0.3. Using Homology, Lemma lemma four lemma (Tag 05QA) again we see that $H^{n-1}(\delta)$ is an isomorphism. The claim holds by induction, so δ is a quasi-isomorphism which is what we wanted to show. $\square$